x_1	x_2	y
1	0	0
2	1	0
3	1	0
4	2	1
5	3	1

$\theta_1 = 0, \ \theta_2 = 0$

$\alpha = 0.1$

α row column

$$\begin{matrix}\theta_1 \\ \theta_2\end{matrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\theta_1 = \theta_1 + 0.1 \left[[0 - 0^{+}] \cdot 1 + \right.$$

$$h_\theta x_1 = \frac{1}{1+e^{-(\theta_1 x_1^1 + \theta_2 x_2^1)}}$$

$$= \frac{1}{1+e^{\theta_1 x_1}} \quad \leftarrow \ \cancel{\theta_2 x_2}$$

$$= \frac{1}{1+e^{0}} \quad \cancel{\theta}$$

$$= \frac{1}{1+1}$$

$$= \frac{1}{2}$$

$$= 0.5$$

$$h_\theta x_2 = \frac{1}{1+e^{-\theta}}(0.5)$$

$$h_\theta x_3 = 0.5$$

$$h_\theta x_4 = 0.5$$

$$h_\theta x_5 = 0.5$$

$$\theta_j = \theta_j + \alpha \sum_{i=1}^{n} (y^i - h_\theta(x^i)) x_j^i$$

$$\theta_1 = \theta_1 + \alpha \left[(y^1 - h_\theta x^1) x_1^1 + (y^2 - h_\theta x^2) x_1^2 \right.$$

$$\theta_1 = \theta_1 + \alpha \left[(0-0.5) \cdot 1 + (0-0.5) 2 + (0-0.5) \cdot 3 + (1 - 0.5) \cdot 4 + (1 - 0.5) \cdot 5 \right]$$

$$\theta_1 = 0 + 0.1 \left[(-0.5) + (-1) + (-1.5) + (+2.0) + (+2.5) \right]$$

$$\theta_1 = 0 + 0.1 [$$

$$\theta_1 = 0 + (-0.75)$$

$$= -0.75$$

$$\theta = 0.15$$

$$\theta_2 = 0 + 0.1 \left[(0-0.5) \cdot 0 + (0-0.5) 1 + (0-0.5) 1 + (1-0.5) 2 + (1-0.5) 3 \right]$$

$$= 0 + 0.1 \left[0 + (-0.5) + (-0.5) + 1 + 1.5 \right]$$

$$= 0 + 0.1 [1.5]$$

$$= 0 + 0.15$$

$$\boxed{\theta_2 = 0.15}$$